

Part 10

MySQL Database in Django

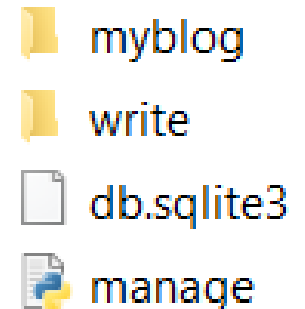
VERSION: 4.0.1

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How to connect MySQL database in Django Project

By default, when we made our first app and started the server you must have seen a new file in your project directory, named as '**db.sqlite3**'. The file is database file where all the data that you will be generating will be stored. It is a local file as Django is a server-side framework and it treats your computer as the host when you actually run the server in command line/terminal.

This file is generated automatically because Django has a default setting of the database set to the SQLite, which is although fine for testing and provides lots of features but if you want your website to be scalable, you need to change it to any other efficient database.



1. DATABASES Dictionary Indexes

Firstly, open the settings.py file of your web-application/ project and there find this part.

```
# Database
# https://docs.djangoproject.com/en/2.1/ref/settings/#databases

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',
        'NAME': os.path.join(BASE_DIR, 'db.sqlite3'),
    }
}
```

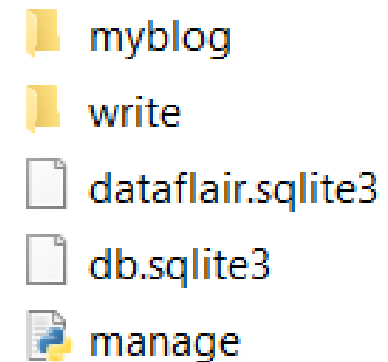
This partition has information regarding the connection to the database.

Create a Migration for SQLite-3

In this, we are also passing the name of the database file or if the file is not present, this will create the db.sqlite3 file. If you change the name to db1.sqlite3 or anything of your choice it will create that file in your root directory every time you run server again.

```
>>> python manage.py migrate
```

```
>>> python manage.py runserver
```



2. Connecting MySQL Database with Django Project

MySQL is a very powerful database providing you with tons of features and flexibility. We are not going to discuss its functionality now, we will just integrate it with our project.

1. Install Xampp:

Xampp is a free opensource tool which provides you with the Apache server and phpMyAdmin which is the best source for beginner programmers to work with MySQL.

You can download the Xampp according to your system from here:

<https://www.apachefriends.org/download.html>

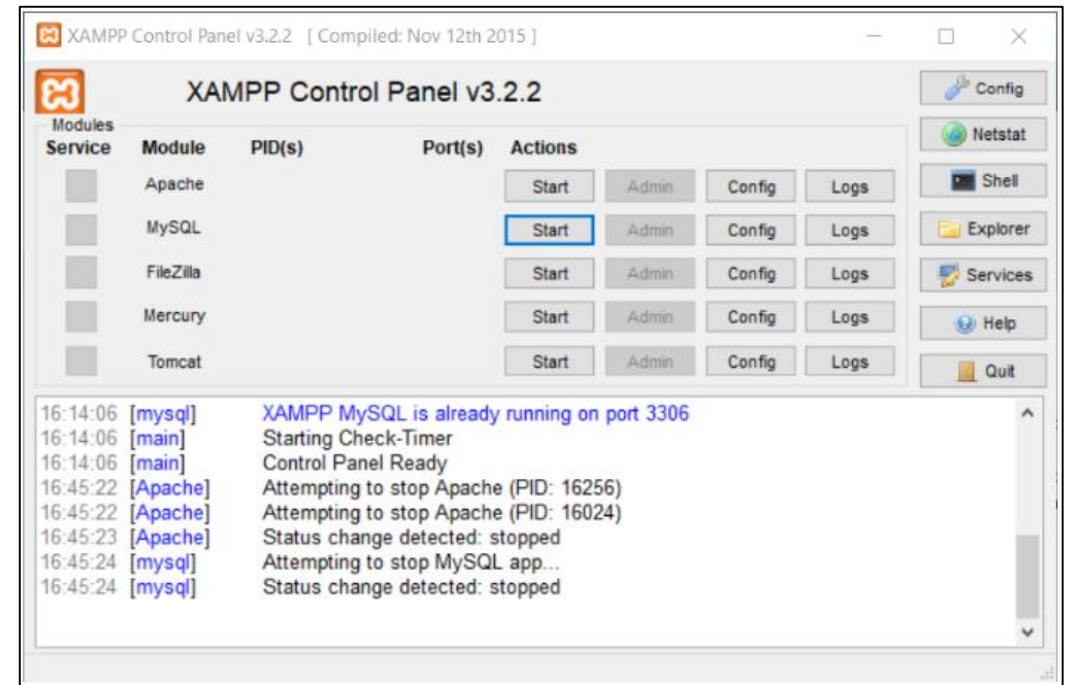
Continue...

2. Run Xampp Control Panel:

Now, after installation, you will have to run the Xampp Control Panel and just start 2 services there, **Apache** and **MySQL**.

Note:

Start the Apache Server First and then the MySQL server.



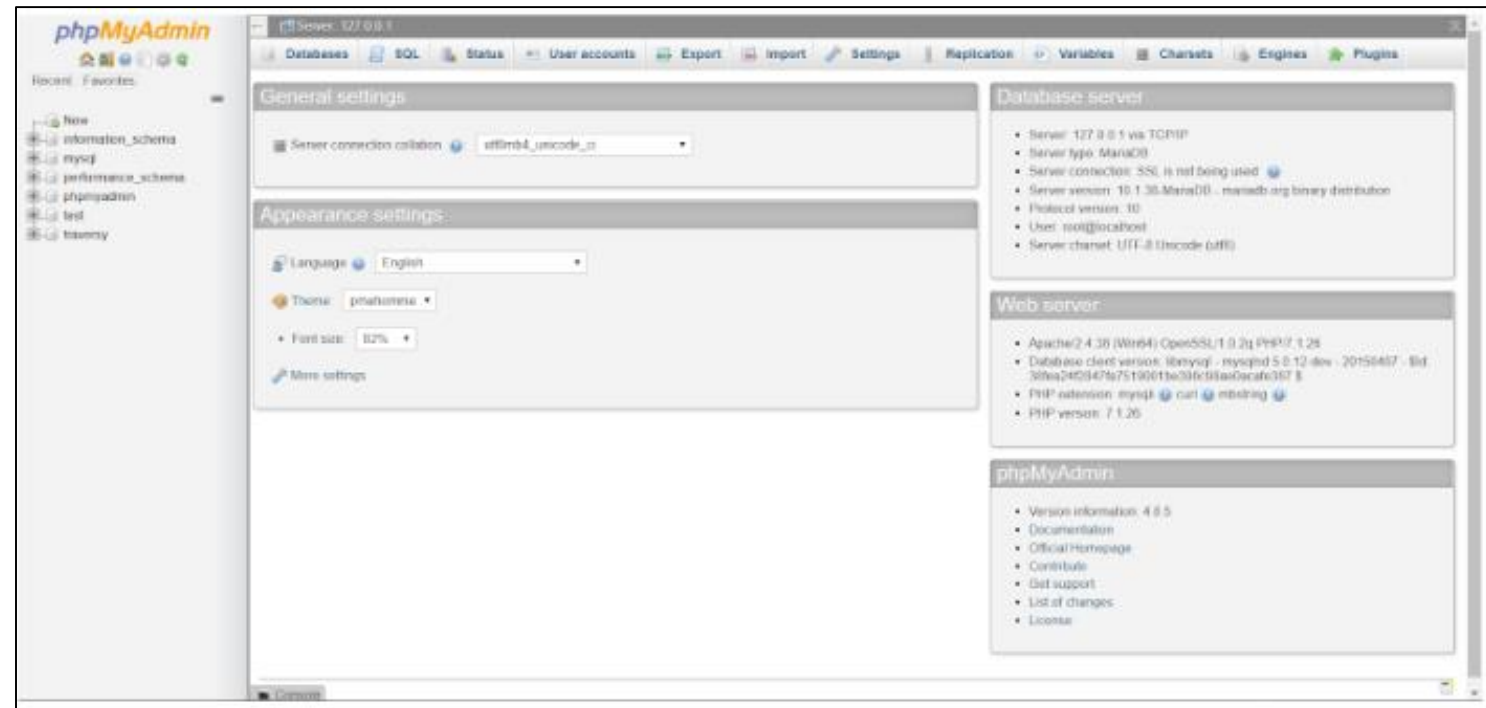
Continue...

Just click on start action, and after starting it should look like this image.

Modules				
Service	Module	PID(s)	Port(s)	Actions
☐	Apache	16256 16024	80, 443	Stop Admin Config Logs
☐	MySQL	9660	3306	Stop Admin Config Logs
☐	FileZilla			Start Admin Config Logs
☐	Mercury			Start Admin Config Logs
☐	Tomcat			Start Admin Config Logs

Continue...

Now, click on the Admin of the MySQL Service, that should open a webpage(offline) looking like this.



Continue...

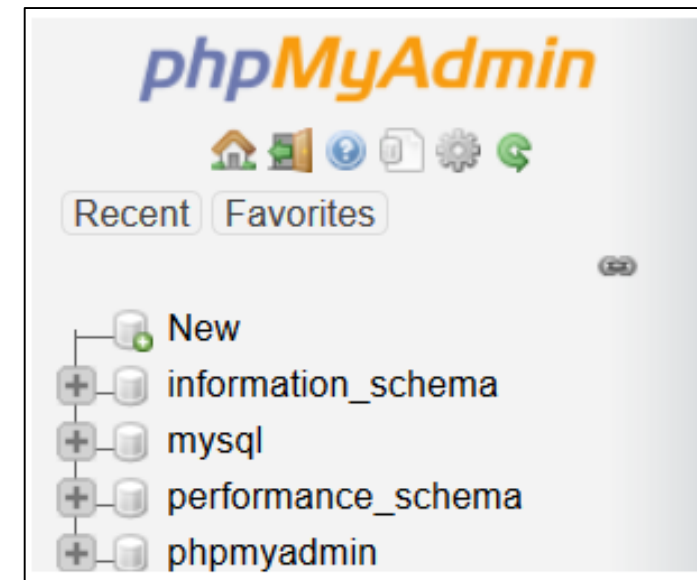
The main advantage of Xampp is that it provides you a very interactive environment and when we will be deploying our models, that too will be done in this.

So, we get a very powerful database with an interactive dashboard and one of the world's best framework with the easiest and most powerful programming language out there.

3. Creating a SQL database

On the webpage phpMyAdmin, we will have to create a database for our project. That's very easy.

Just Click on the New button as shown here. Then, just fill the desired name of your database and click on create button.



4. Modifying settings.py

In this last step, we will be changing the DATABASE dictionary in our main projects settings.py.

First, install this file via command line:

>>> pip install mysqlclient

```
DATABASES = {  
    'default': {  
        'ENGINE': 'django.db.backends.mysql',  
        'NAME': 'dataflair',  
        'USER': 'root',  
        'PASSWORD': '',  
        'HOST': '',  
        'PORT': '',  
        'OPTIONS': {  
            'init_command': "SET sql_mode='STRICT_TRANS_TABLES'"  
        }  
    }  
}
```

It will install the Django code for connecting the MySQL Database.

Continue...

For the last steps just run these two commands:

```
>>> python manage.py migrate
```

```
>>> python manage.py runserver
```

Now, refresh the phpMyAdmin page and you will get some tables created with just doing this much.

That's it, you have now all the things ready for your website, all the components, and development setup is finally complete.

Summary

We learned how to add the MySQL database to our project and also understood the DATABASE dictionary of the Django Framework.

Also, we learned that Django has middleware and libraries for almost all the important databases out there. And, you won't need to learn a new database language for that, Django does all that work for you.

Thanks for Reading